# Formula 1 Podium Prediction

Milestone 1 analytical report

## 1. Overview

The task is to predict whether a driver finishes on the podium (`positionOrder <= 3`) using only information available after qualifying and before the race starts. One row is one driver in one race. Results, points, laps, status, pit stops, lap times and the standings published for the same race are not used as features.

The selected model is a random forest. On the unseen test seasons (2022–2024) it reaches PR-AUC 0.715, ROC-AUC 0.938 and F1 0.683, against PR-AUC 0.647 for a simple grid rule and 0.150 for chance.

## 2. Data cleaning and EDA

- **Sentinels and types.** `\N` was replaced with missing values; dates, numbers and lap-time strings were typed (0 conversion failures). In `constructor_results.status`, `\N` means “not disqualified” and was coded as such.
- **Keys.** 39 of 42 key checks passed before cleaning and 42 of 43 after. The lap-time key was repaired: 1,710 exact copies dropped and 269 conflicting copies resolved and logged.
- **Duplicates.** 85 driver-race pairs (176 rows, 1950–1978, mostly shared drives) were collapsed by keeping the best-classified stint. The alternative rule (most laps) keeps the same row in all 85 cases. Result: 27,499 rows, one per (raceId, driverId).
- **Coverage.** Qualifying starts in 1994 and is complete only from 2003, so modeling uses 2003–2024. A grid of 0 means “no grid slot” and never produces a podium.

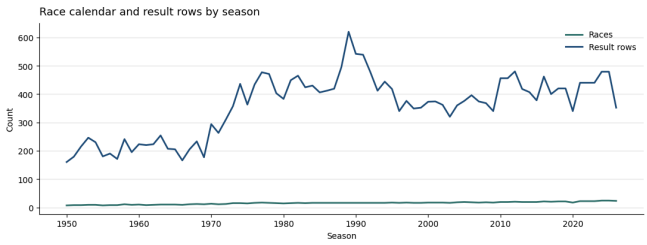



Figure 1. Races and result rows by season.

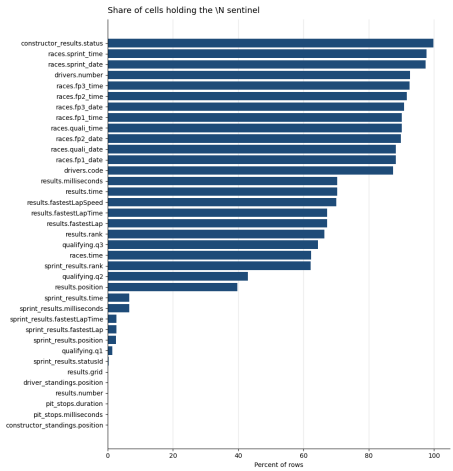



Figure 2. Share of `\N` sentinels by column.

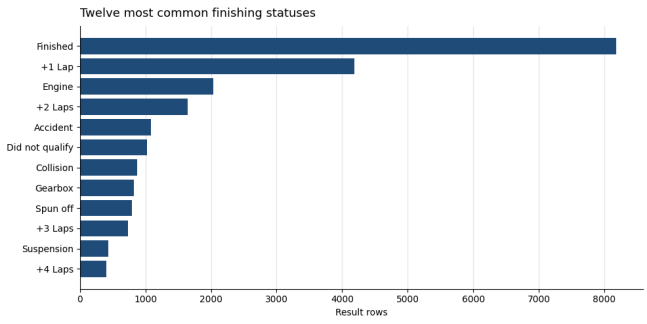



Figure 3. Most common finishing statuses.

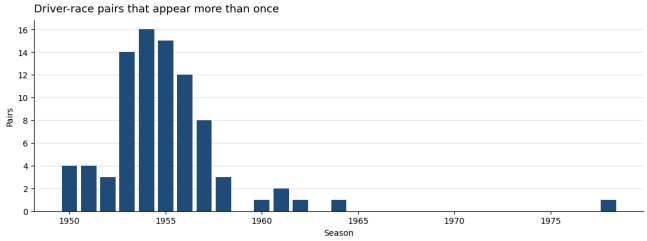



Figure 4. Duplicate driver-race pairs by season.

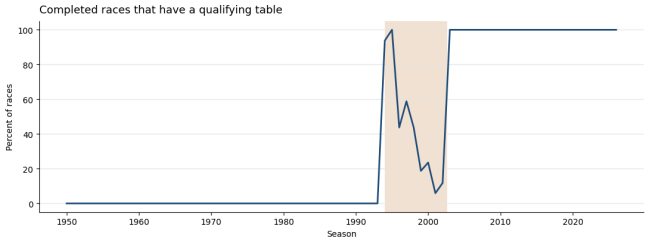



Figure 5. Races with qualifying data by season.

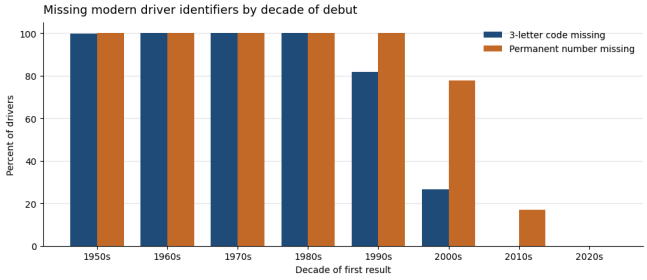



Figure 6. Missing driver codes and numbers by debut decade.

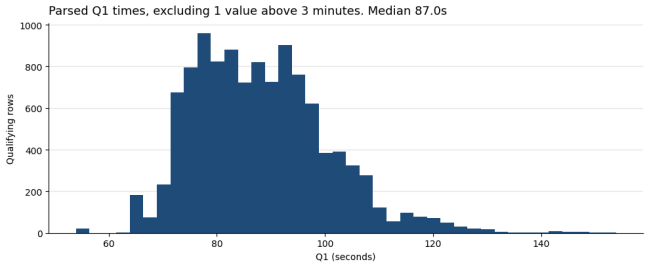



Figure 7. Parsed Q1 times in seconds.

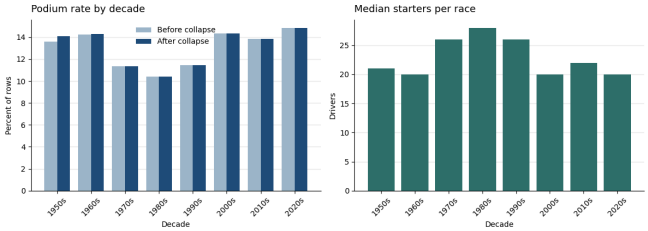



Figure 8. Podium rate by decade before and after the duplicate collapse.

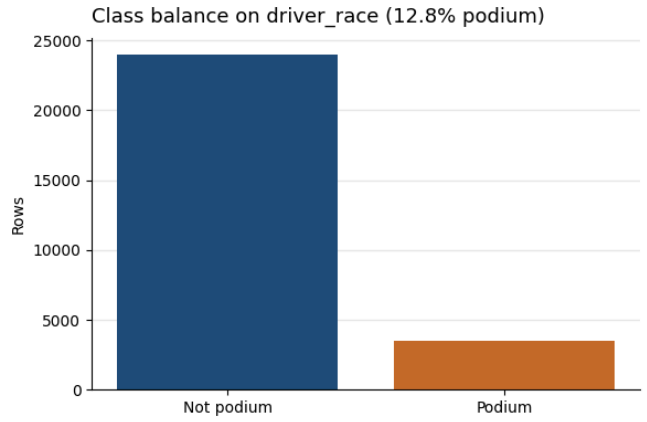



Figure 9. Class balance: 12.8% podium.

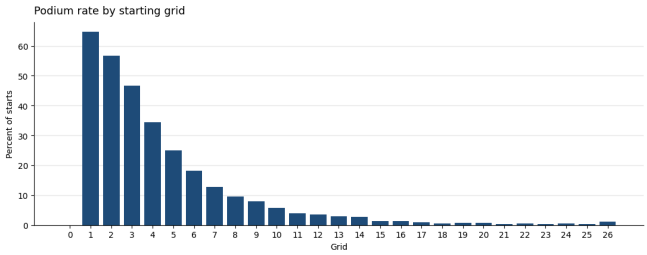



Figure 10. Podium rate by starting grid.

# 3. Data-engineering questions

### Q1. Which circuits convert a front-row start into a podium most often, and does this change after 2014?

**Method.** driver_race → races → circuits, with qualifying for comparison. One row per start from grid 1 or 2; circuits need at least 20 such starts per era. The starting grid is used rather than the qualifying position, because penalties separate them: drivers who qualified on the front row and started there podiumed 72.6% of the time, against 28.9% for those moved back.

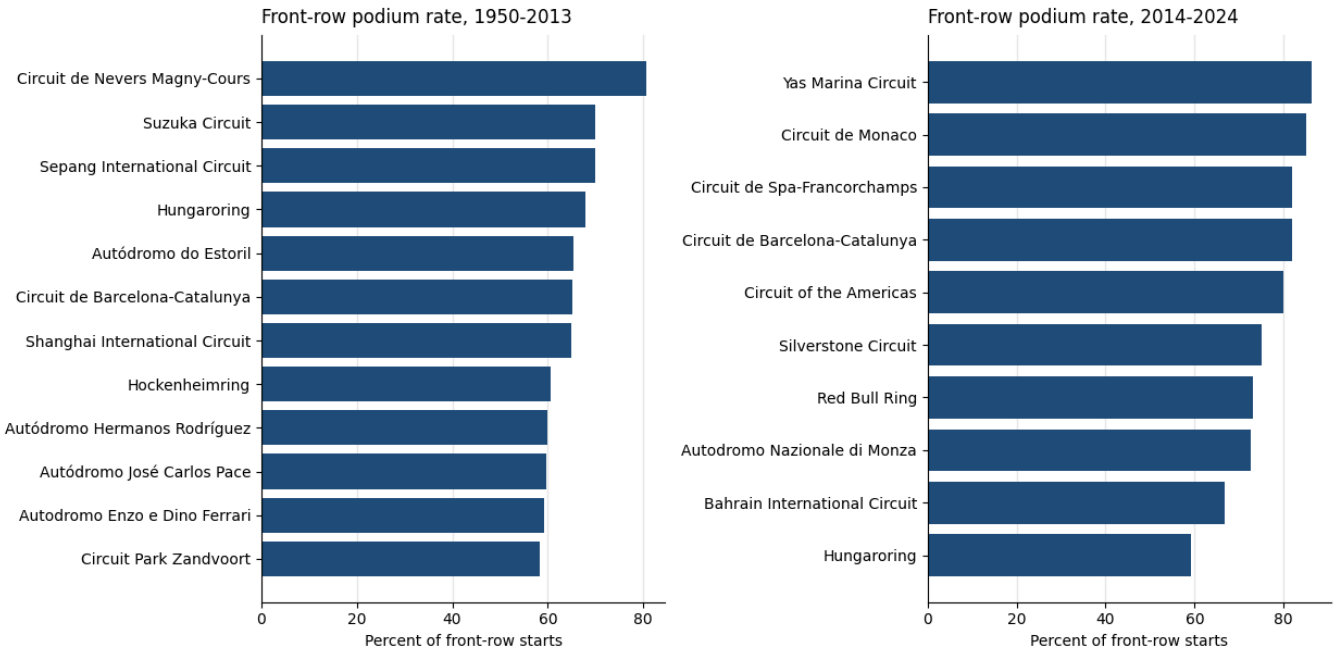


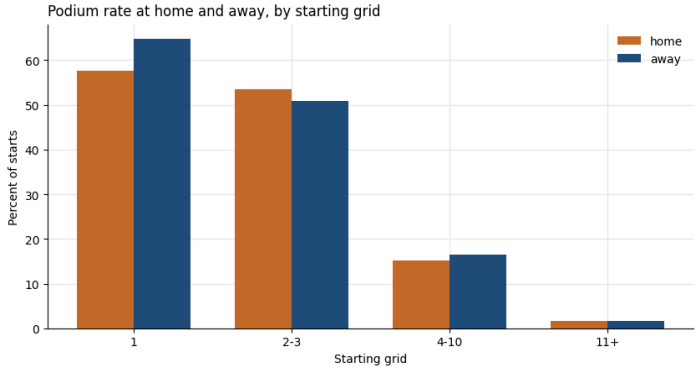

Figure 11. Front-row podium rate by circuit, 1950–2013 and 2014–2024.

| 1950–2013 | Rate | 2014–2024 | Rate |
|---|---|---|---|
| Magny-Cours | 80.6% (29/36) | Yas Marina | 86.4% (19/22) |
| Sepang | 70.0% (21/30) | Monaco | 85.0% (17/20) |
| Suzuka | 70.0% (35/50) | Barcelona | 81.8% (18/22) |
| Hungaroring | 67.9% (38/56) | Spa | 81.8% (18/22) |
| Estoril | 65.4% (17/26) | Circuit of the Americas | 80.0% (16/20) |

**Answer.** Yes, partly. Front-row conversion is higher overall from 2014 (76% of 454 front-row starts, against 68% of 400 in 2003–2013). Before 2014 Magny-Cours leads; from 2014 Yas Marina and Monaco lead. Individual circuits rest on only 20–26 front-row starts since 2014, so a gap of 5 points is about one podium and close ranks are within noise. Only the large moves stand out: Monaco rises from 23rd to 2nd and Monza from 25th to 8th, while the Hungaroring drops from 4th to 10th.

### Q2. Do drivers podium more often at home, after controlling for starting position?

**Method.** driver_race → drivers (nationality) and races → circuits (country). Home and away rates are compared inside grid bands (pole, 2–3, 4–10, 11+). Rows with no grid slot are excluded.




| Grid | Home | Away | Gap |
|---|---|---|---|
| Pole | 57.6% (92) | 64.7% (1,032) | −7.1 |
| 2–3 | 53.5% (172) | 51.0% (2,074) | +2.5 |
| 4–10 | 15.1% (649) | 16.6% (7,231) | −1.5 |
| 11+ | 1.6% (1,411) | 1.7% (12,372) | 0.0 |

Figure 12. Podium rate at home and away by grid band.

**Answer.** No. Within each grid band the home rate is not higher; the weighted gap is −0.53 points and no band differs by more than 1.33 standard errors.

### Q3. Which constructors have the highest mechanical-retirement rate in 2014–2021 vs 2022–2024?

**Method.** driver_race → status → constructors. The denominator is race starts; entries that never started are removed. A mechanical retirement is a status naming a failed component; crashes, driver errors, regulation decisions and the generic "Retired" are excluded. Every constructor with at least 30 starts in a period is ranked for that period. A renamed team has a new constructorId in the source (Toro Rosso → AlphaTauri, Renault → Alpine, Racing Point → Aston Martin), so each name is ranked in the period it raced under. The full list of statuses and their categories is in the notebook (Section 15).

| Period | Highest rate | Second | Whole field |
|---|---|---|---|
| 2014–2021 | Caterham, 23.5% (8 of 34 starts) | Lotus F1, 20.0% | 10.1% |
| 2022–2024 | Alfa Romeo, 12.5% (11 of 88 starts) | Alpine, 11.0% | 6.1% |

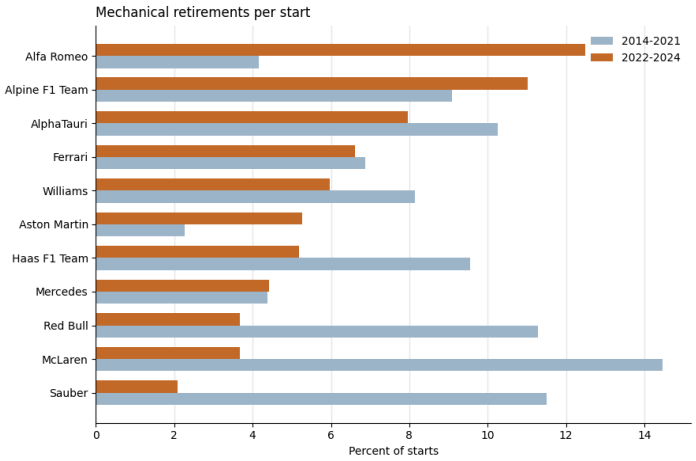



Figure 13. Mechanical retirements per start, constructors with at least 30 starts in both periods.

| Constructor | 2014–21 | 2022–24 |
|---|---|---|
| McLaren | 14.5% | 3.7% |
| Sauber | 11.5% | 2.1% |
| Red Bull | 11.3% | 3.7% |
| AlphaTauri | 10.3% | 8.0% |
| Haas | 9.5% | 5.2% |
| Alpine | 9.1% | 11.0% |
| Williams | 8.2% | 6.0% |
| Ferrari | 6.9% | 6.6% |
| Mercedes | 4.4% | 4.4% |
| Alfa Romeo | 4.2% | 12.5% |
| Aston Martin | 2.3% | 5.3% |

**Answer.** Among every constructor with at least 30 starts in the period, the highest mechanical-retirement rate in 2014–2021 is Caterham at 23.5% (8 of 34 starts), ahead of Lotus F1 at 20.0%. In 2022–2024 it is Alfa Romeo at 12.5% (11 of 88 starts), ahead of Alpine at 11.0%. Across the whole field the rate falls from 10.1% to 6.1% of starts. Among the 11 constructors present in both periods (Figure 13), the largest fall is McLaren (14.5% → 3.7%) and the largest rise is Alfa Romeo (4.2% → 12.5%). Caterham's rate rests on only 34 starts, where one failure moves it by about 3 points, so the order at the top is uncertain. Alpine and Aston Martin's 2014–2021 rates cover 2021 only, and Sauber's 2022–2024 rate covers 2024 only.

## 4. Selected features

The 16 features fall into five groups, chosen from the training-year podium rates in Figure 14. Each is known before the race starts.

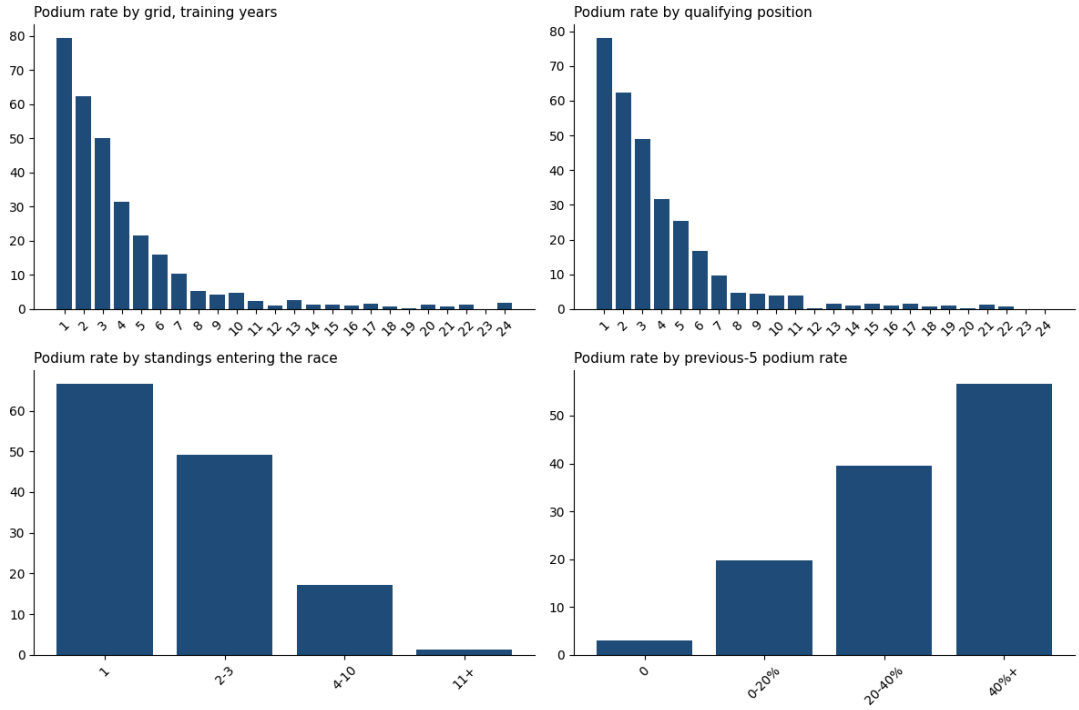



Figure 14. Training-year podium rate by grid, qualifying position, entering standing and recent form.

| Feature | Source | Why it is known before the race |
|---|---|---|
| `grid`, `front_row`, `no_grid_slot` | results.grid | Starting grid is published before the start |
| `quali_position`, `reached_q3`, `quali_position_missing` | qualifying | Qualifying ends before the race |
| `driver_podium_rate_last5`, `driver_points_last5` | results, previous 5 starts | Earlier races only; current race excluded |
| `constructor_podium_rate_last5` | results, previous 5 team races | Earlier races only |
| `career_starts`, `form_missing` | results, earlier starts | Count of past starts |
| `driver_standing_entering`, `constructor_standing_entering`, `standing_missing` | standings at round − 1 | Published after the previous round |
| `season_round` | races.round | Fixed by the calendar |
| `home_race` | drivers + circuits | Static facts |

Excluded: the current race’s result, points, laps, status, times, lap times, pit stops, standings and same-weekend sprint results.

## 5. Split and models

The split is chronological: train 2003–2018 (6,411 rows), validation 2019–2021 (1,200), test 2022–2024 (1,359). Hyperparameters and thresholds were chosen on validation only. The primary metric is PR-AUC because podiums are rare (14.0% of training rows, 15.0% of validation and test rows); the secondary metric is F1 at the validation-chosen threshold.

**Models.** The *grid rule* predicts a podium for a top-three start and is the floor every model must clear. *Logistic regression* (baseline) combines the imputed, standardised features in a weighted sum passed through the logistic function; it is simple and stable, but it has one weight per feature and cannot capture interactions, such as a pole start being worth more in a front-running car. The *random forest* (selected) averages 300 trees of depth at most 8, each grown on a bootstrap sample with a random subset of features at each split; it captures interactions and barely overfits (train 0.746, validation 0.749 PR-AUC), but its probabilities are not calibrated and correlated features such as grid and qualifying position share importance. *Histogram gradient boosting* adds shallow trees one at a time, each correcting the errors of the trees before it; it is the most flexible family and overfits here (train–validation PR-AUC gap 0.06 for its best setting, up to 0.18 for others). The *shallow neural network* (two hidden layers of 32 and 16 units, early stopping) learns non-linear combinations of the features, but it needs scaled inputs, depends on its random seed and a random early-stopping hold-out, is the hardest to explain, and does not beat the forest on about 6,400 training rows.

| Model | PR-AUC val / test | ROC-AUC val / test | F1 val / test | Precision val / test | Recall val / test |
|---|---|---|---|---|---|
| Grid rule (grid ≤ 3) | 0.676 / 0.647 | 0.917 / 0.914 | 0.717 / 0.649 | 0.717 / 0.650 | 0.717 / 0.647 |
| Logistic regression (baseline) | 0.729 / 0.710 | 0.931 / 0.938 | 0.725 / 0.683 | 0.649 / 0.597 | 0.822 / 0.799 |
| **Random forest (selected)** | **0.749 / 0.715** | **0.937 / 0.938** | **0.733 / 0.683** | **0.665 / 0.613** | **0.817 / 0.770** |
| Gradient boosting | 0.721 / 0.684 | 0.936 / 0.913 | 0.729 / 0.635 | 0.719 / 0.658 | 0.739 / 0.613 |
| Shallow NN (32, 16) | 0.737 / 0.706 | 0.932 / 0.932 | 0.725 / 0.665 | 0.692 / 0.654 | 0.761 / 0.676 |

Performance on the training years, at the same validation-chosen thresholds:

| Model | PR-AUC | ROC-AUC | F1 | Precision | Recall |
|---|---|---|---|---|---|
| Grid rule (grid ≤ 3) | 0.618 | 0.907 | 0.638 | 0.638 | 0.638 |
| Logistic regression (baseline) | 0.684 | 0.928 | 0.665 | 0.577 | 0.784 |
| **Random forest (selected)** | **0.746** | **0.947** | **0.697** | **0.628** | **0.783** |
| Gradient boosting | 0.779 | 0.954 | 0.710 | 0.697 | 0.722 |
| Shallow NN (32, 16) | 0.703 | 0.935 | 0.667 | 0.640 | 0.698 |

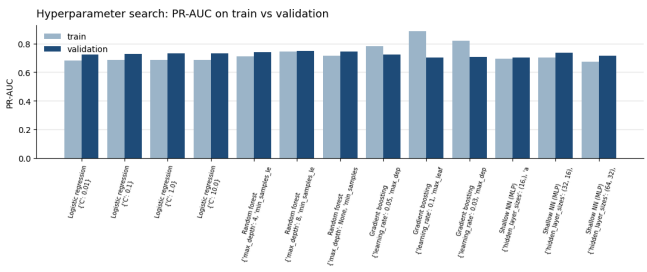



Figure 15. Train vs validation PR-AUC for every setting tried.

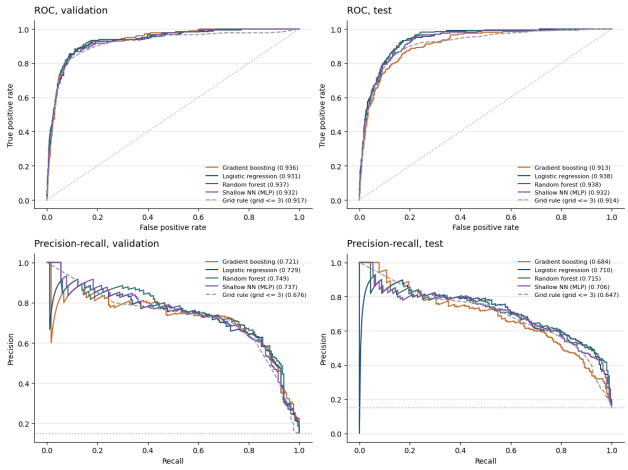



Figure 16. ROC and precision-recall curves.

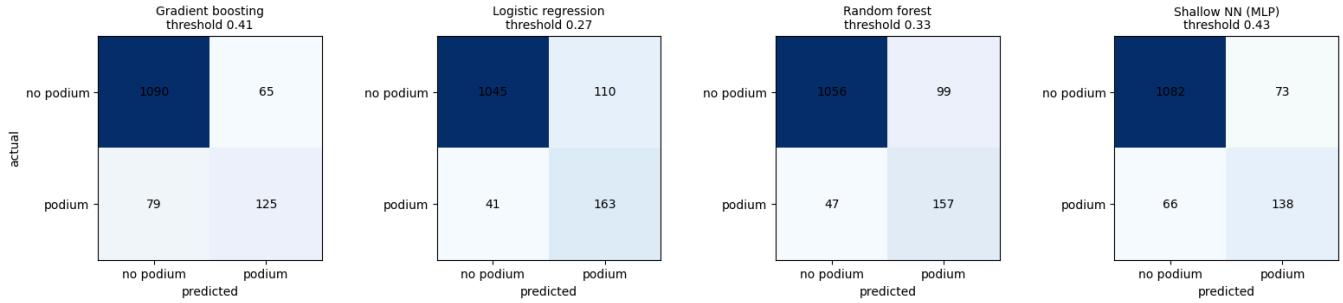



Figure 17. Test confusion matrices at the validation thresholds.

## 6. Ablation and preprocessing order

Removing one feature group at a time and refitting shows that grid matters most (validation PR-AUC drop 0.034), then qualifying (0.017) and standings (0.009). Form adds almost nothing (0.001) and home slightly hurts (−0.002).

Three preprocessing-order experiments keep the model fixed and swap only the order of two steps. Two of them are flat on the random forest by design. Clipping to the 5th–95th percentiles leaves the median unchanged, so clipping before or after median imputation produces identical data for any model. A tree depends only on the order of the values, so a quantile transform before or after imputation gives the same splits. The experiment that can move is on the logistic baseline: scaling before or after imputation changes validation PR-AUC by only 0.0002. The order of these steps therefore does not matter for this data. The notebook (Section 18) also repeats the clipping and quantile orders on logistic regression.

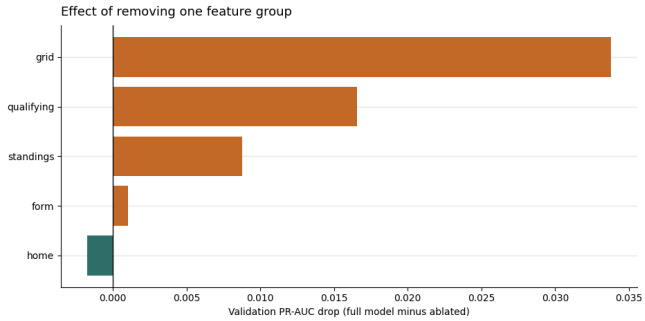



Figure 18. Validation PR-AUC drop per removed group.

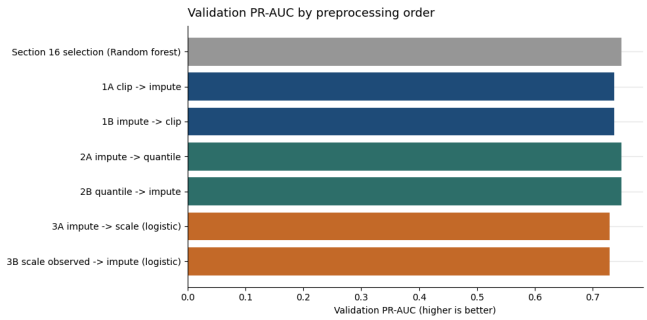



Figure 19. Validation PR-AUC by preprocessing order.

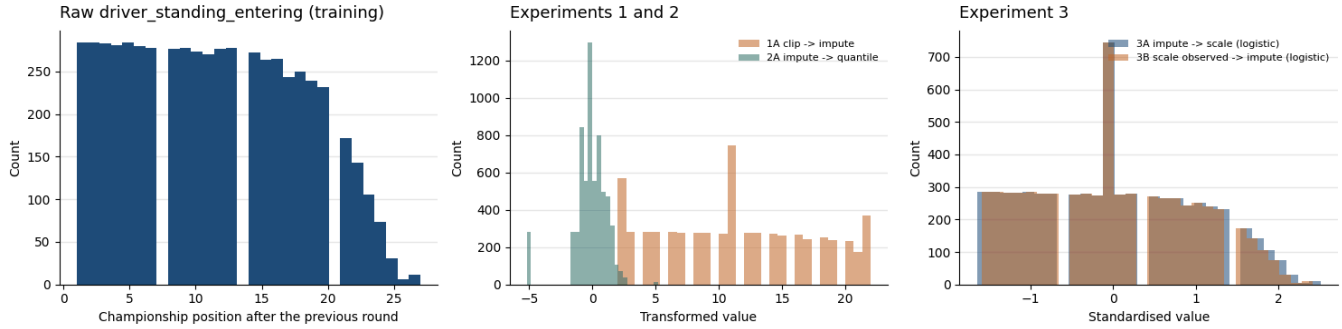



Figure 20. Effect of each preprocessing step on one feature.

## 7. Explainability

Grid and qualifying position dominate both permutation importance and SHAP. The entering standings and recent form come next in SHAP, but their permutation importances are within their spread across shuffles (form 0.006 ± 0.013), and removing the form group costs only 0.001 PR-AUC, because form, standings and grid carry overlapping information.

The SHAP plots use 40 validation rows; the notebook (Section 19) repeats the global SHAP on all 1,200 validation rows. For one validation row (Verstappen, 2021 Austrian GP, predicted 86.1%), SHAP and LIME both attribute the prediction mainly to pole position, P1 in qualifying and recent podiums. These explanations show what the model relies on; they are not causal effects.

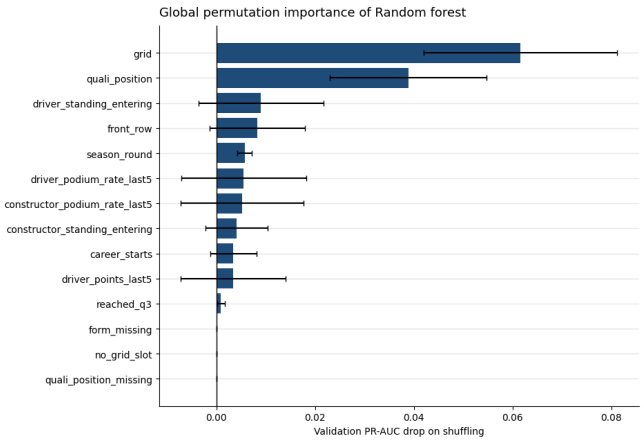



Figure 21. Permutation importance.

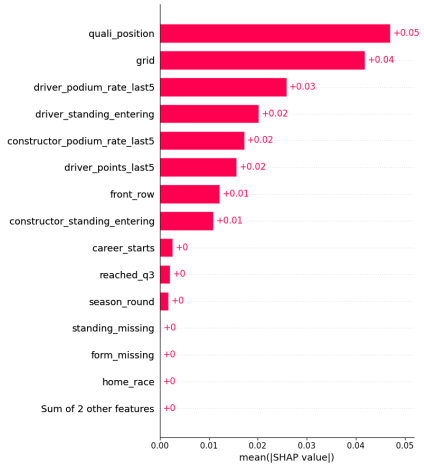



Figure 22. Mean |SHAP| (40 validation rows).

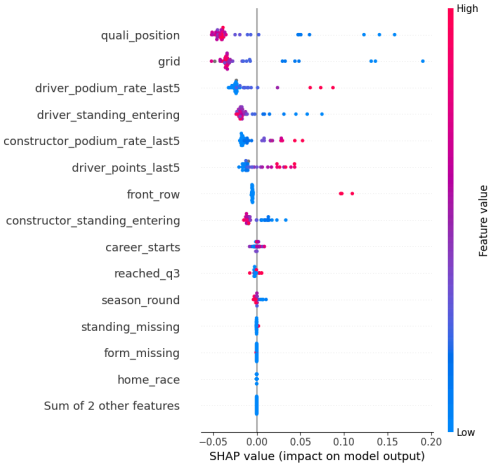



Figure 23. SHAP beeswarm (40 validation rows).

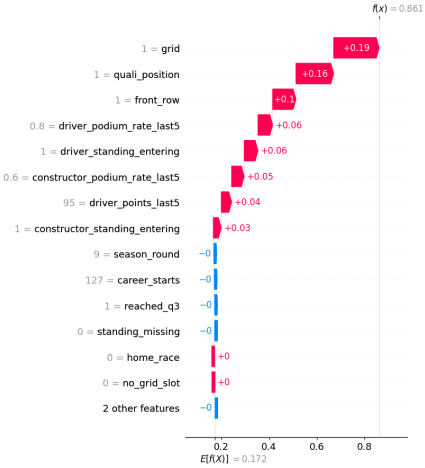



Figure 24. Local SHAP for one prediction.

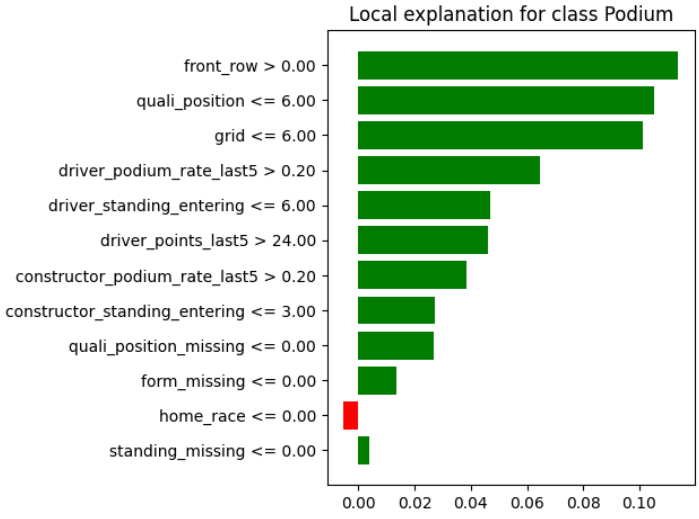



Figure 25. LIME for the same prediction.

## 8. Inference function

`predict_podium(record)` takes the pre-race fields for one driver-race, refuses post-race fields, and returns the podium probability, the decision at the validation threshold (0.333) and the missing fields. Missing fields are filled with training medians.

| Record | Missing fields | P(podium) | Call | Actual |
|---|---|---|---|---|
| Complete (Verstappen, 2022 Canadian GP) | none | 0.881 | podium | podium |
| Partially missing (same entry) | qualifying and both standings | 0.557 | podium | podium |

## 9. Known limitations

1. Qualifying data starts in 1994 and is complete only from 2003, so the model applies to 2003–2024 only.
2. Era sensitivity: the model was trained on 2003–2018; held-out PR-AUC varies from 0.670 to 0.779 by season (Figure 26), and 2022–2024 is a new rules era.
3. Missing inputs are rare in training, so a missing field is filled as a midfield value and lowers the prediction (0.881 → 0.557 above).
4. Only 17 pit-lane starts in training, so the model learned little from them.
5. Team renames reset constructor form (42 rows).
6. On 2021–2022 sprint weekends the grid differs from qualifying for 77–82% of drivers.
7. Q1 rankings rest on 20–26 front-row starts per circuit, so close ranks are within noise.
8. Q2 depends on matching nationality to circuit country.
9. Q3 excludes the generic “Retired” status, so mechanical rates are a lower bound; a team with few starts (Caterham, 34) can top the list on a handful of failures.
10. Predicted probabilities are not calibrated, and a race can get more or fewer than three podium calls.
11. `driver_points_last5` mixes points systems: a win was worth 10 points until 2009 and 25 from 2010, so the same value means different form in the early training years and in the test years.
12. Q3 ranks each constructorId, so a renamed team counts as a different constructor in each period.

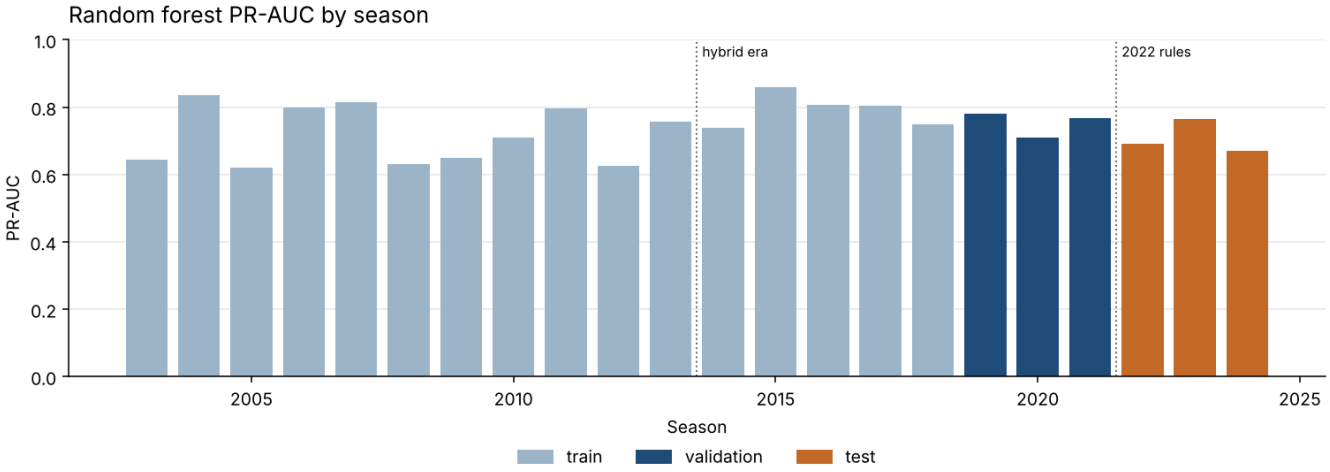



Figure 26. Random forest PR-AUC by season.